

United International University

Department of Computer Science and Engineering

Final Examination Spring 2025

Course Code: **CSE 1112** Course Title: **Introduction to Computer Systems**

Date: July 11, 2025 (Tuesday) Time: 09:00 AM – 10:00 AM (1 hour) Full marks: 20

Name:

Student ID:

Write down C programs for the following problems in Code Blocks (or any C compiler you prefer), and present the code to your instructor after the time is up. You can make rough calculations in this paper. **Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.**

Question 1: You are a software engineer developing a Hogwarts advising platform. Your task is to implement a program that calculates and adjusts a student's semester cost based on their ID and credit hours.

Inputs (3): Mark 3

1. **Student ID:** An multi- digit integer
2. **Theory Credit Hours:** Integer
3. **Lab Credit Hours:** Integer

Task 1: Basic Semester Cost Calculation Mark 3

- Theory course cost per credit: **10,000.00 BDT**
- Lab course cost per credit: **8,000.00 BDT**

Output 1: Print the **total semester cost**

Task 2: Theory Discount Mark 5

Apply a **30% discount on the theory course cost only if:**

- Theory credit hour = sum of the **last two digits** of the student ID

Output 2: Print the **discounted theory cost**

Task 3: Lab Discount

Mark 3

Apply a **20% discount on the lab course cost only if:**

- Lab credit hour is divisible by **either the last digit or the second last digit** of the student ID

Output 3: Print the **discounted lab cost** as a float.

Sample Input:	Sample Output:
Student ID: 20233457	Total Semester Cost: 168000.00
Theory Credit: 12	Discounted Theory Cost: 84000.00
Lab Credit: 6	Discounted Lab Cost: 38400.00

Question 2: A stadium has a circular shape with a rectangular field. Write a C program to: **(Mark 6)**

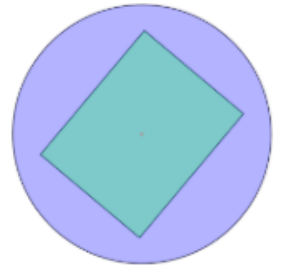
1. Calculate the total stadium area (treat it as a circle):

$$\text{stadium_area} = \pi \times r \times r$$

2. Calculate the field area (rectangle):

$$\text{field_area} = \text{height} \times \text{width}$$

3. Calculate the seating area, which is the difference between the stadium and the field: $\text{seating_area} = \text{stadium_area} - \text{field_area}$



Sample Input:	Sample Output:
• Enter r_squared: 100 • Enter height (a): 8 • Enter width (b): 10 Use $\pi = 3.1416$. Print all areas rounded to 2 decimal places.	Stadium Area: 314.16 Field Area: 80.00 Seating Area: 234.16